EMPOWERING INDIA : ANALYSING THE EVOLUTION OF UNION BUDGET ALLOCATION FOR SUSTAINABLE GROWTH

1.Introduction:

Project Title : Empowering India: Analysing the Evolution of Union Budget Allocations for Sustainable Growth

The Union Budget of India is one of the most important policy documents of the Government, outlining its financial strategy and development priorities for a given fiscal year. It presents a comprehensive account of the government's estimated revenues and planned expenditures, reflecting its approach towards economic growth, social welfare, and national development. Through strategic budgetary allocations, the government seeks to address key challenges such as poverty reduction, infrastructure development, employment generation, and sustainable economic progress.

1.1 Project Overview:

Title: Empowering India: Analysing the Evolution of Union Budget Allocations for Sustainable Growth

Objective:

- To analyze the changes in Union Budget allocations over multiple financial years.
- To examine sector-wise distribution of funds to identify development priorities.
- To evaluate how budget allocations contribute to sustainable growth and national development

Expected Outcomes :

- Clear understanding of trends and changes in Union Budget allocations across financial years.
- Identification of key sectors receiving priority funding for sustainable growth.
- Data-driven insights presented through visualizations to support policy and academic analysis

1.2 Purpose:

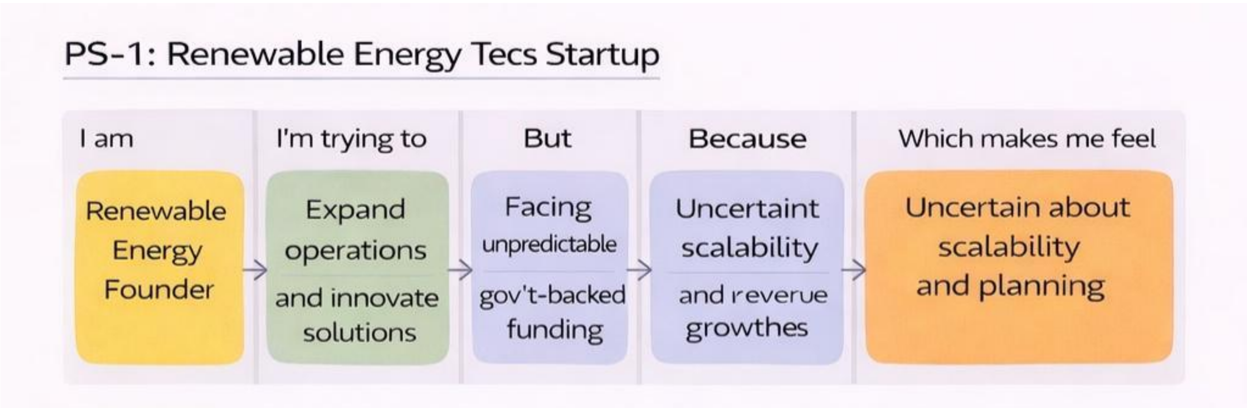
- To examine how Union Budget allocations influence India's economic and social development.
- To identify trends and priority sectors in government spending over the years.
- To provide analytical insights that support understanding of sustainable growth strategies.

2.Ideation Phase

2.1 Problem Statement:

Despite the Union Budget being a key instrument for India's economic and social development, there is limited consolidated analysis that clearly explains how budget allocations have evolved over the years and how effectively they support sustainable growth. The absence of structured, visual, and comparative analysis makes it difficult for students and stakeholders to understand sector-wise priorities, spending trends, and their long-term impact on empowering India. This project addresses this gap by systematically analyzing Union Budget allocations across multiple financial years to derive meaningful, data-driven insights.

Example :



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Renewable Energy Founder	Expand operations & innovative solutions	Facing unpredictable gov't backed funding	Uncertain scalability and revenue growth	Uncertain about scalability and planning

2.2 Empathy Map Canvas:

The Empathy Map for Empowering India: Analyzing the Evolution of Union Budget Allocation for Sustainable Growth focuses on policy-aware citizens, researchers, and development professionals who seek to understand how India's fiscal priorities shape long-term economic, social, and environmental outcomes. They think critically about whether Union Budget allocations truly align with sustainable development goals and feel both hopeful and skeptical about the government's commitment to inclusive and green growth

Brain Storming:

In the project "Empowering India: Analyzing the Evolution of Union Budget Allocations for Sustainable Growth," the brainstorming phase focused on analyzing and visualizing Union Budget data (2021–2024) to support business and policy decisions. Key ideas included studying budget allocations across ministries, departments, schemes, and categories, identifying top-funded schemes, and comparing revenue and capital expenditures.

.Step-1: Team Gathering, Collaboration and Select the Problem Statement

TEAM

R Yamuna

T Neeraja

T Madhulatha

U Madhavi

Union Budget data of India is large and complex, making it difficult to understand how funds are allocated across ministries and schemes. This project converts raw budget data into simple and interactive Tableau dashboards to help users identify trends and support better decision-making for sustainable growth.

Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

- **10 minutes** to prepare
- **1 hour** to collaborate
- **2-8 people** recommended

Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

10 minutes

A Team gathering
Define who should participate in the session and send an invite. Share relevant information or pre-work ahead.

B Set the goal
Think about the problem you'll be focusing on solving in the brainstorming session.

C Learn how to use the facilitation tools
Use the Facilitation Superpowers to run a happy and productive session.

Open article →

1 Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

5 minutes

PROBLEM

How might we [your problem statement]?

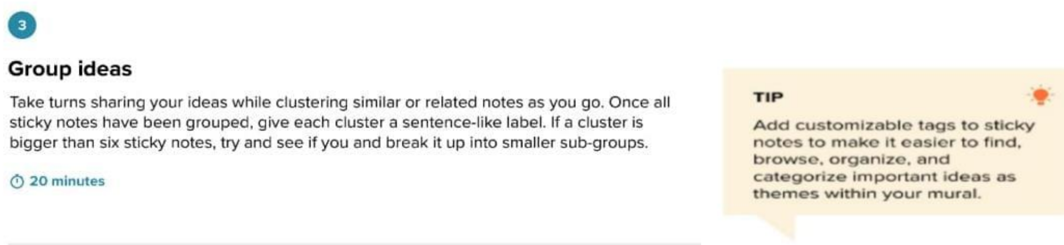
Union Budget data is complex and vast. Empowering India simplifies it using SQL and Tableau.Clear visualizations reveal spending trends.It Supports informed decisions for sustainable growth.

Key rules of brainstorming

To run an smooth and productive session

- Stay in topic.
- Encourage wild ideas.
- Defer judgment.
- Listen to others.
- Go for volume.
- If possible, be visual.

Step-2: Brainstorm, Idea Listing and Grouping



Cluster-1: Budget Trends & Growth Analysis

- Year-wise budget comparison (2021–24)
- Budget growth trends
- Revenue vs Capital analysis
- High-growth sector identification

Cluster-2: Ministry & Department Insights

- Top ministries by allocation
- Sustainable development focus
- Allocation changes analysis
- Department-wise distribution

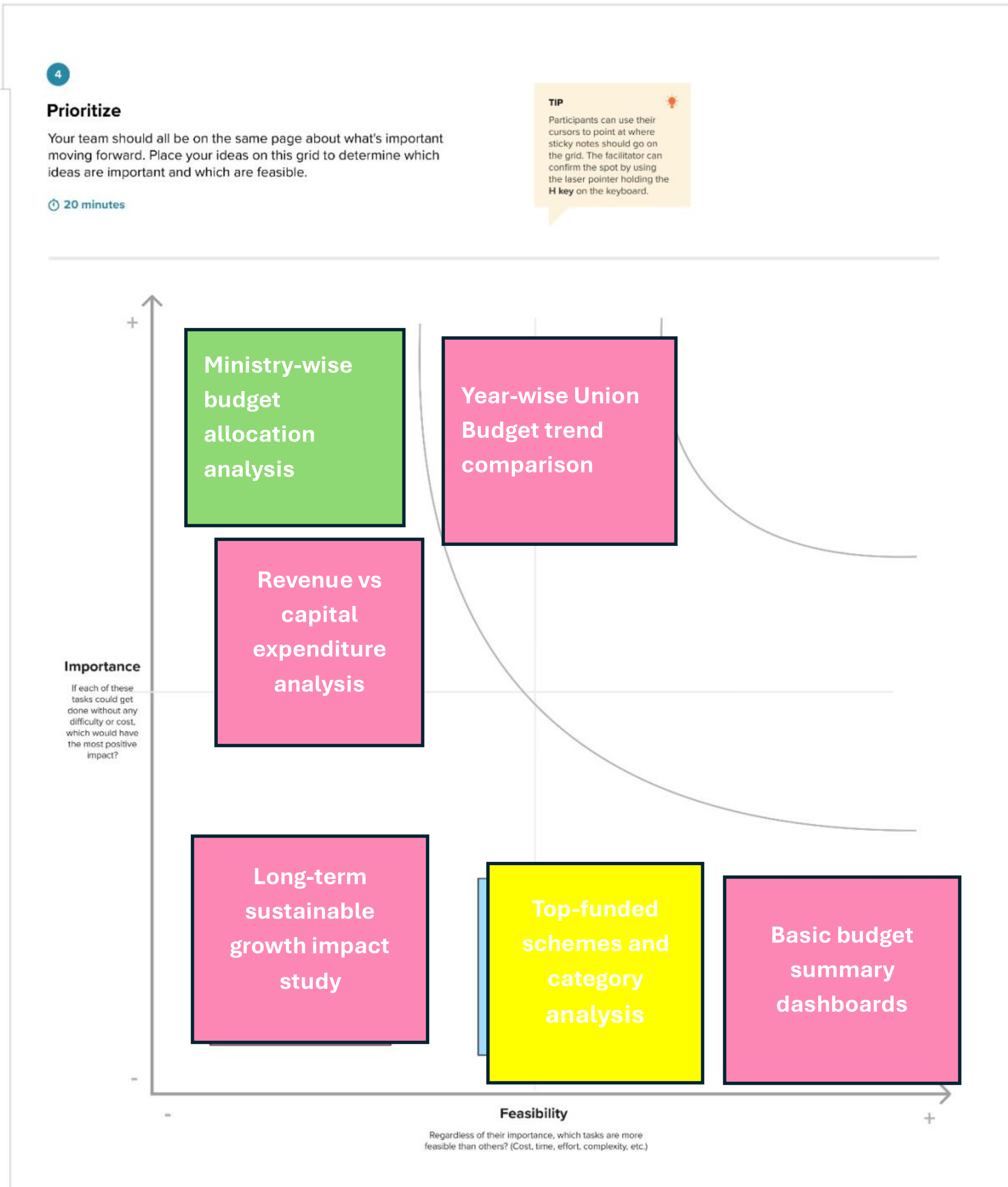
Cluster-3: Scheme & Category Analysis

- Scheme-wise allocation study
- Top-funded schemes
- Major budget. changes
- Category-wise fund distribution

Cluster-4: Visualization & Decision Support

- Interactive Tableau dashboards
- Data-driven decision support
- Insightful Tableau stories
- Filters (year, ministry, scheme)

Step-3: Idea Prioritization



3. Requirement Analysis

3.1 Customer Journey Map:

Scenario	Entice	Enter	Engage	Exit	Extend
Steps	User discovers dashboard through project presentation / link	Opens Tableau Public / Server dashboard	Filters data (year, sector, ministry) and interacts with charts	Interprets trends and downloads insights	Shares insights in report / presentation
Steps	Understand budget allocation trends	Access dashboard quickly and easily	Analyze sector-wise and year-wise allocations	Extract meaningful conclusions	Use insights for academic / business decisions
Goals	Understand budget allocation trends	Access dashboard quickly and easily	Analyze sector-wise and year-wise	Easy export option	Data supports decision making
Positive Experience	Clear project description	Fast loading dashboard	Interactive filters and clean visuals	Easy export option	Data supports decision making
Negative Experience	Hard to find dashboard link	Poor layout / too many charts	Confusing filters or cluttered visuals	No clear summary provided	No follow-up or advanced insights

3.2 Solution Requirement:

Functional Requirements : .

FR No	Functional Requirement (Epic)	Sub Requirement (Story/Sub-Task)
FR-1	Data Collection & Integration	Collect and integrate Union Budget data from multiple years
FR-2	Data Cleaning & Transformation	Clean, validate, and standardize budget datasets
FR-3	Budget Allocation Analysis	Analyze sector-wise and year-wise allocation trends
FR-4	Dashboard & Visualization	Create interactive dashboards with filters
FR-5	Report & Insights	Generate and export analytical reports
FR-6	Sustainable Growth Indicators	Map budget data to sustainable indicators
FR-7	Historical Trend Analysis	Store and analyze historical budget data
FR-8	Predictive Analysis	Forecast future budget allocation trends
FR-9	User Access & Security	Ensure secure and controlled access

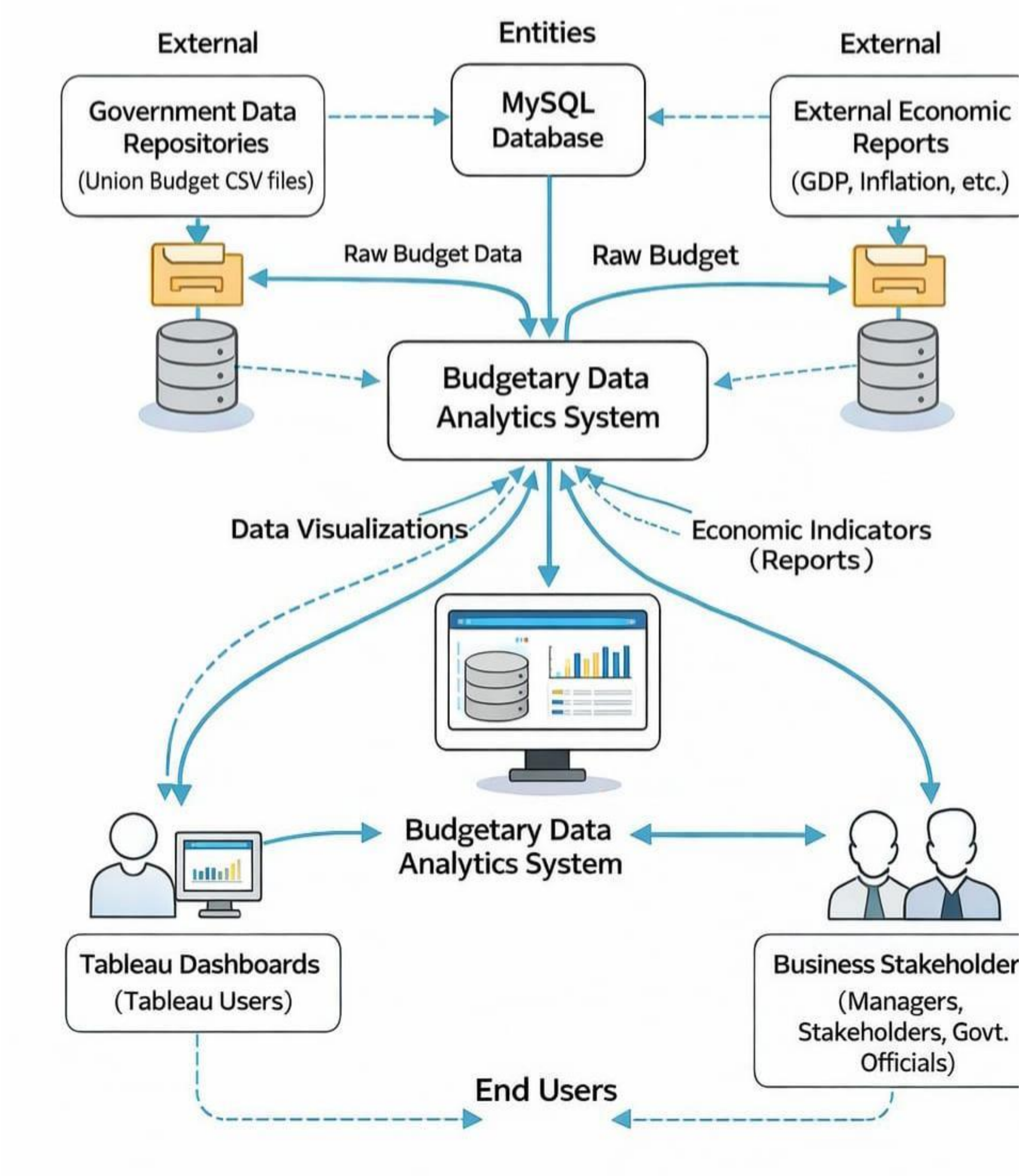
Non-Functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No	Non-Functional Requirements	Description
NFR-1	Usability	The Tableau dashboards and reports should be Intuitive and easy for various user types to navigate and understand.
NFR-2	Security	User access to dashboards and underlying data must be authenticated and authorized based on roles.
NFR-3	Reliability	The data ingestion and transformation processes should reliably complete daily refreshes without errors, ensuring consistent data availability.
NFR-4	Performance	Dashboards should load and render within 5-10 seconds for typical usage, even with large datasets.
NFR-5	Availability	The Tableau Server and underlying data sources should be available 99.5% of the time during business hours .
NFR-6	Scalability	The system must be able to handle a 50% increase in data volume and a 30% increase in concurrent users over the next 2 years without significant performance degradation.
NFR-7	Data Integrity	All data loaded into the system must be accurate, consistent, and complete, matching the source systems after transformation.
NFR-8	Maintainability	The data pipelines, Tableau workbooks, and server configurations should be well-documented and easily maintainable by the IT team.
NFR-9	Extensibility	The system should be capable of integrating new data sources (e.g., social media data, loT production data) with minimal development effort.
NFR-10	Response Time	Complex data queries and dashboard filter applications should return results within 15 seconds.

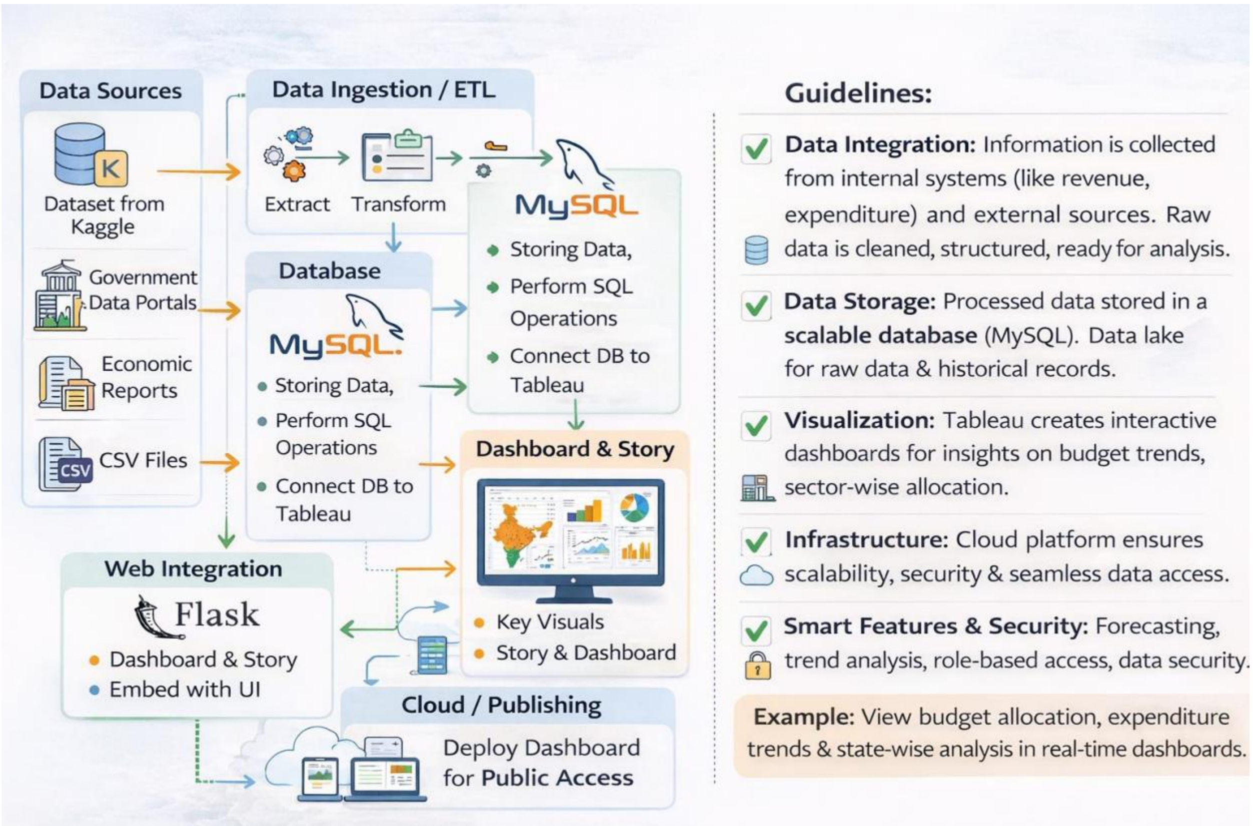
3.3 Data Flow Diagram

"Empowering India: Analyzing the Evolution of Union Budget Allocations for Sustainable Growth," data is collected from sources like government portals, Kaggle datasets, and CSV files. This data is cleaned and combined to remove errors and make it ready for analysis. The processed data is stored securely in a database so it can be easily accessed and analyzed. Business intelligence tools like Tableau use this data to create interactive dashboards and charts that show trends and insights clearly..



3.4 Technology Stack

The technology architecture defines the structured framework that explains how data flows through the system, from collection to final visualization. It begins with multiple data sources such as government portals, Kaggle datasets, CSV files, and economic reports. This data is processed using ETL (Extract, Transform, Load) techniques to clean, integrate, and prepare it for analysis.



4. Project Design Phase

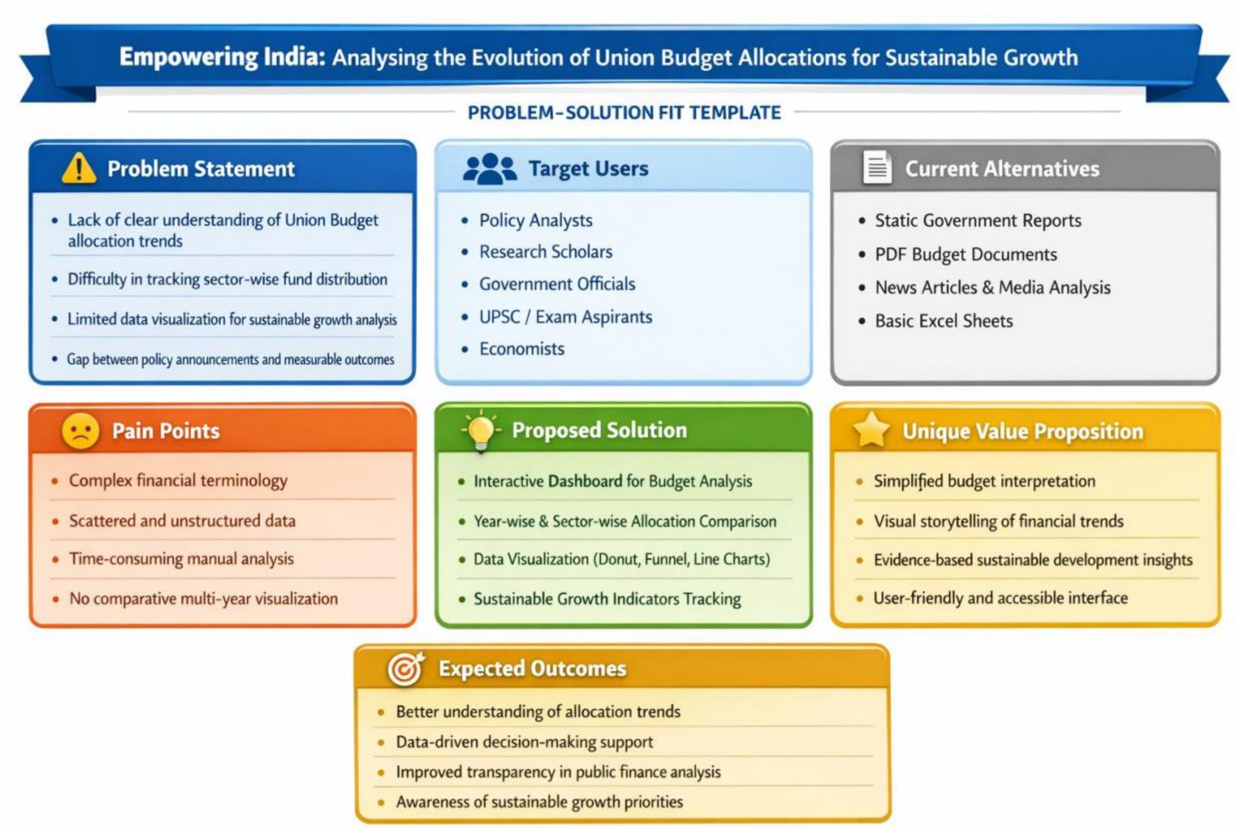
4.1 Problem Solution Fit:

Problem-Solution Fit :

The proposed Union Budget Analysis Dashboard achieves strong problem–solution fit by addressing the difficulty in understanding complex and scattered budget data from 2021–2024. Many users struggle to analyze sector-wise allocations and year-wise trends due to lack of clear visualization. The SQL + Tableau dashboard simplifies this data into interactive and easy-to-understand visuals.

Purpose:

- Validates that the solution is solving a real and clearly defined problem.
- Ensures alignment between user needs and project features.
- Reduces the risk of developing unnecessary or irrelevant functionalities.
- Improves the overall effectiveness and quality of the solution.



4.2 Proposed Solution:

S.No	Parameter	Description
1	Problem Statement	Government budget data from 2021–2024 is complex and difficult to analyze due to large datasets and multiple sectors. It is hard to track allocation trends, sector growth, and spending patterns clearly
2	Idea / Solution Description	Develop an Interactive SQL + Tableau dashboard that integrates Union Budget data (2021–2024) into a centralized system to visualize sector-wise allocation, revenue trends, and expenditure analysis.
3	Novelty / Uniqueness	Combines structured SQL data processing with interactive Tableau storytelling to provide clear insights on budget trends, comparisons, and growth patterns across years
4	Social Impact	Improves transparency and awareness of government spending. Helps students, researchers, and policymakers understand budget priorities and economic impact
5	Business Model	Can be offered as a research tool for institutions or as a consulting dashboard for policy analysis.
6	Scalability of the Solution	Easily scalable to include future budget years, state budgets, or ministry-level deep analysis using cloud-based data storage and Tableau dashboards

4.3 Solution Architecture:

1. **Data Source Layer :**
    - Collects official Union Budget data for financial years 2021–2024.
    - Includes ministry-wise and sector-wise allocation details.
    - Covers revenue, capital expenditure, and total budget amounts.
    - Data gathered from government reports and open data portals.
2. **Data Ingestion Layer :**
    - Extracts raw data from CSV, Excel, and government reports.
    - Cleans and pre-processes data (remove duplicates, handle missing values, format columns).
    - Transforms data into structured format suitable for database storage.
    - Loads processed data into MySQL tables using import tools / SQL commands.
    - Ensures data validation and consistency before moving to analysis layer.
3. **Data Storage Layer :**
    - Stores cleaned and structured budget data in MySQL Database.
    - Organizes data into tables like Year, Ministry, Sector, and Allocation.
    - Maintains relationships using primary keys and foreign keys.
    - Ensures data integrity, consistency, and secure storage.
    - Provides structured data access for SQL queries and Tableau visualization
4. **Visualization & Reporting Layer:**
    - Connects Tableau to MySQL database for real-time data access.
    - Creates interactive dashboards with charts, KPIs, and filters (Year, Ministry, Sector).
    - Visualizes trends, comparisons, and budget distribution clearly.
    - Generates summary reports and downloadable insights.
    - Helps users analyze data and support academic or business decision-making.

5.Project Planning & Scheduling

5.1 Planning Logic :

A Sprint is a fixed period during which a team works to complete a set of tasks. An Epic is a large objective (e.g., full analysis & dashboard/report) broken into smaller Stories. Story Points represent the relative effort required to complete a task (1, 2, 3, 5…).

Sprint 1 (5 Days)

Epic – Data Collection & Preparation

Task	Description	Story Points
Data Collection	Collecting Union Budget data from government reports (2021–2024), CSV, and official portals	3
Data Cleaning	Removing inconsistencies, handling missing values, correcting formats	3
Data Transformation	Structuring data year-wise, sector-wise, and scheme-wise	2
Exploratory Analysis	Understanding trends, major sectors, and allocation patterns	1

Sprint 1 Total Story Points = 9

Sprint 2 (5 Days)

Epic – Analysis & Visualization Development

Task	Description	Story Points
KPI Identification	Identifying key indicators like sector growth, % allocation, YoY changes	2
Visualization Design	Designing charts (bar, line, pie, trend analysis)	3
Dashboard / Report Development	Creating interactive dashboards or structured analytical reports	5
Validation & Accuracy Check	Ensuring correctness of calculations and insights	2

Sprint 2 Total Story Points = 12

Sprint 3 (5 Days)

Epic – Insights, Presentation & Deployment

Task	Description	Story Points
Insight Generation	Deriving meaningful Insights on sustainable growth & sector priorities	3
Storyboard Creation	Structuring insights Into a logical narrative for presentation	2
Stakeholder Review	Collecting feedback and suggested improvements	3
Final Refinements	Polishing visuals, content, and conclusions	2
Tableau Public Deployment	Publishing dashboards online	2

Sprint 3 Total Story Points = 12

Velocity :

- Total Story Points = 9+12+12=33
- Total Sprints=3
- Velocity =33/3=11.0 Story Points as per Sprint

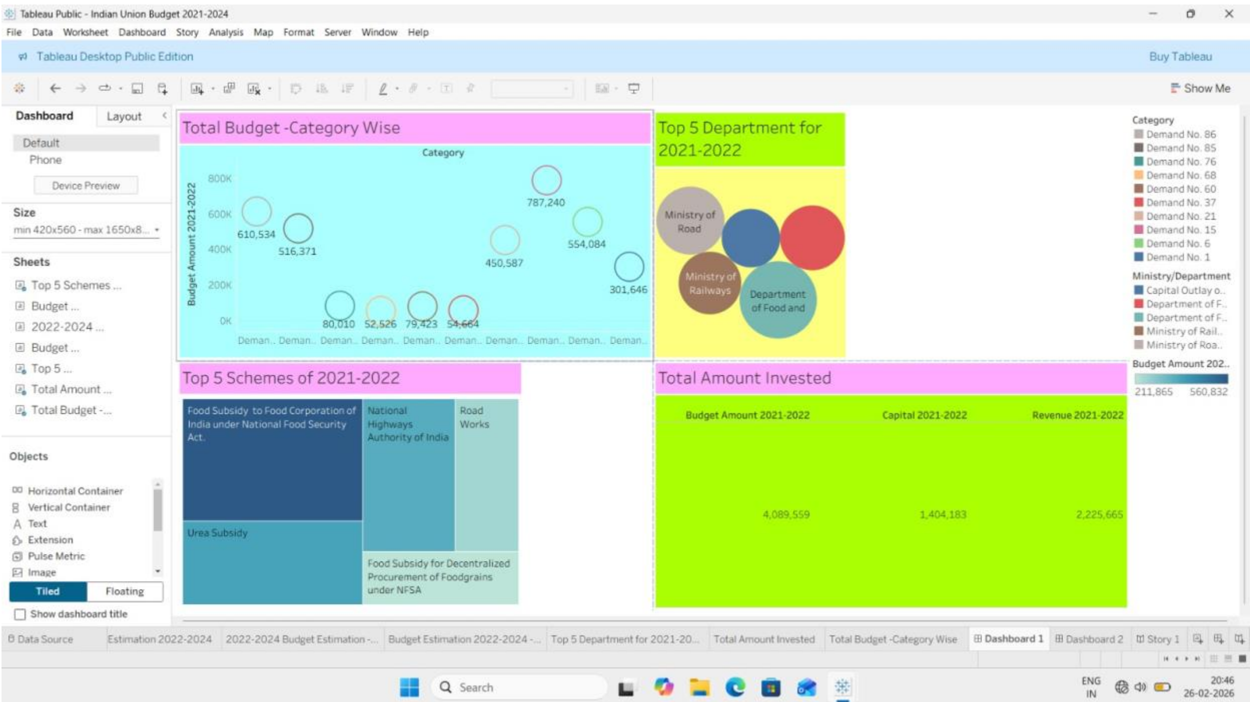
6. Functional & Performance Testing

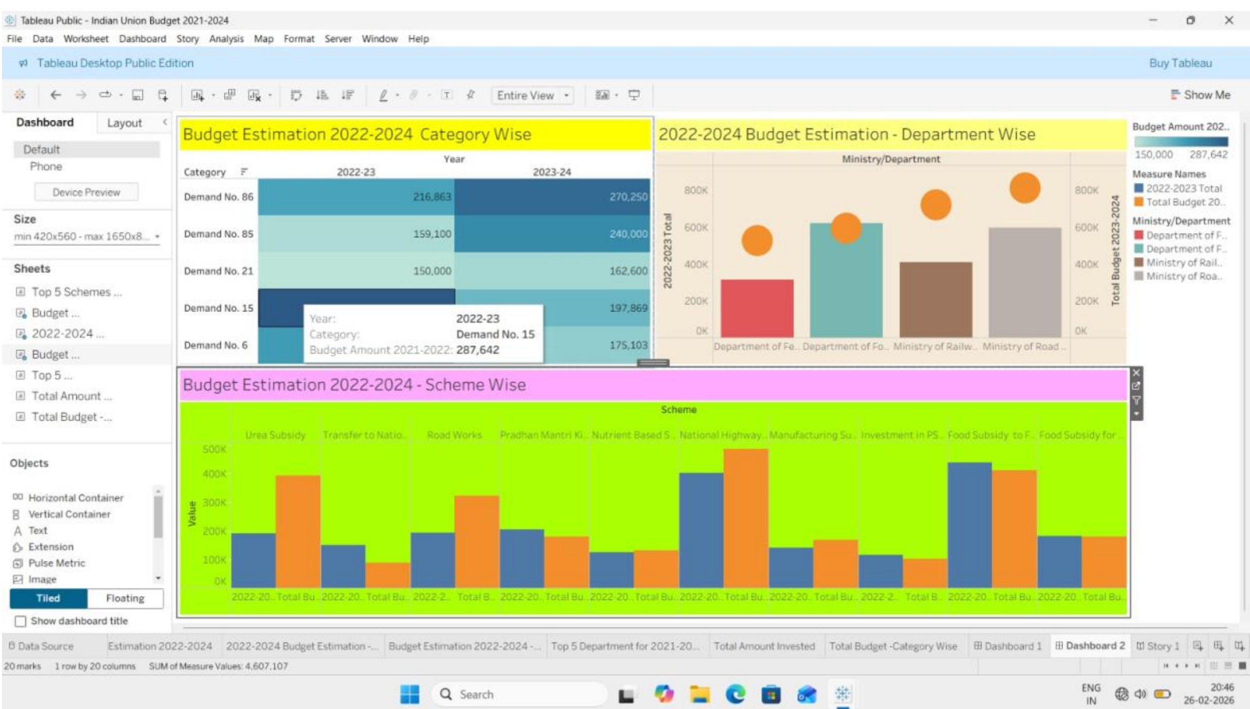
Model Performance Testing :

S.No	Parameter	Screenshot/Value
1	Data Rendered	Displayed Union Budget allocation data across multiple years, sectors, and ministries. Data is rendered using tables and visual charts to represent year-wise and sector-wise budget distribution clearly.
2	Data Preprocessing	Removed missing and inconsistent values, standardized numerical formats (₹ Crores), renamed columns for clarity, and structured data by Year → Sector → Scheme hierarchy.
3	Utilization of filters	Filters applied for Financial Year, Sector, Ministry, Scheme Type, and Allocation Category (Revenue / Capital), enabling dynamic exploration of budget trends
4	Calculation fields Used	Growth Rate (%) = (Current Year Allocation − Previous Year Allocation) / Previous Year Allocation × 100. Percentage Share = (Sector Allocation / Total Budget) × 100.
5	Dashboard Design	No. of Visualizations / Graphs – 7 (Bar Chart, Line Trend Graph, Pie Chart, Sector-wise Comparison Chart, KPI Cards, Heat Map).
6	Story Design	No. of Visualizations / Graphs – 5 (Overall Budget Overview, Sector Growth Analysis, Sustainable Development Focus, Year-wise Trend Analysis, Key Insights & Policy Impact).

7. Results

7.1 Output Screenshots:





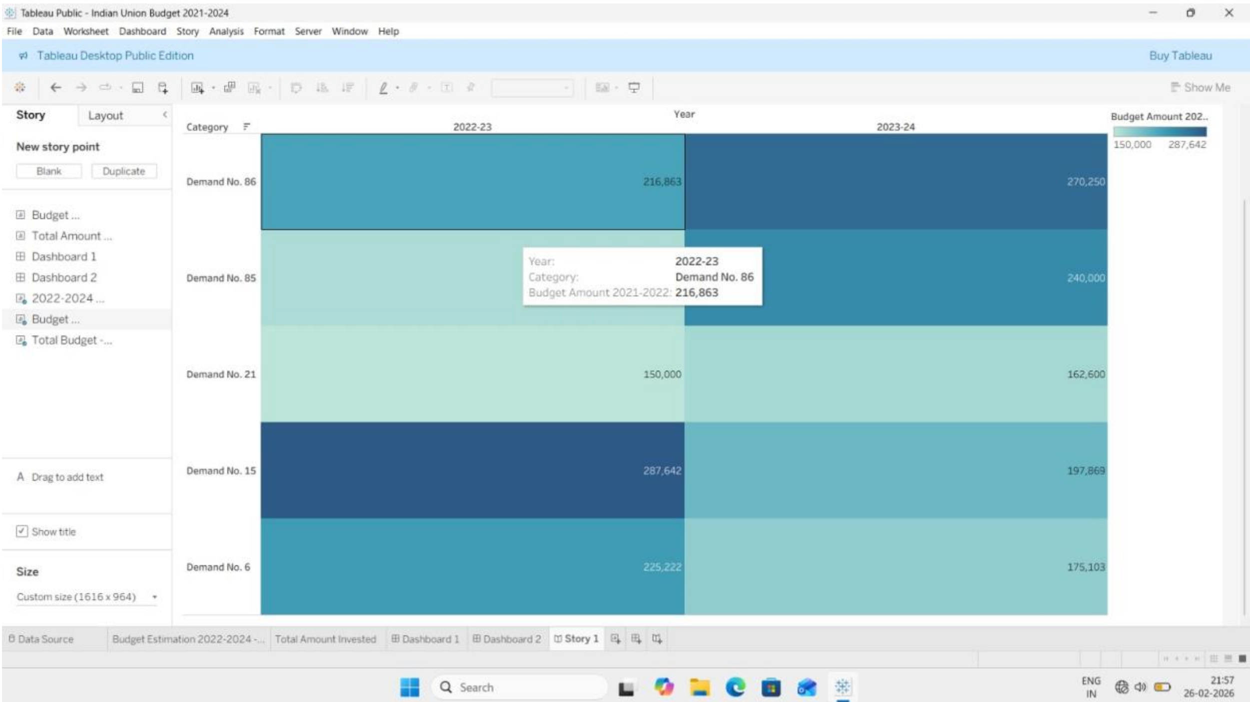


Tableau Public - Indian Union Budget 2021-2024
Tableau Desktop Public Edition
Story
Layout
New story point
Category
Year
2022-23
2023-24
Budget Amount 202..
150,000
287,642
Demand No. 86
Demand No. 85
Demand No. 21
Demand No. 15
Demand No. 6
216,863
270,250
240,000
150,000
162,600
287,642
197,869
225,222
175,103
Year: 2022-23
Category: Demand No. 86
Budget Amount 2021-2022: 216,863
Data Source
Budget Estimation 2022-2024 -...
Total Amount Invested
Dashboard 1
Dashboard 2
Story 1

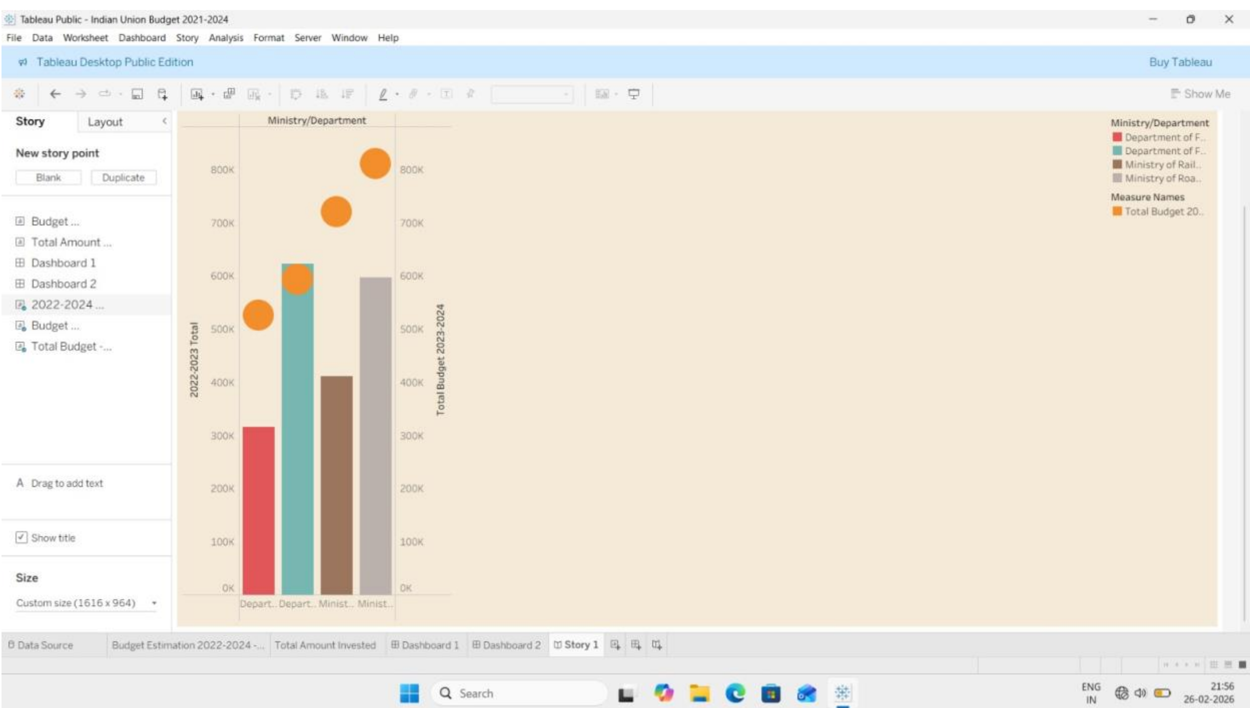


Tableau Public - Indian Union Budget 2021-2024
Tableau Desktop Public Edition
Ministry/Department
2022-2023 Total
Total Budget 2023-2024
Department of F...
Department of F...
Ministry of Rail...
Ministry of Roa...
Measure Names
Total Budget 20..
Depart..
Depart..
Minist..
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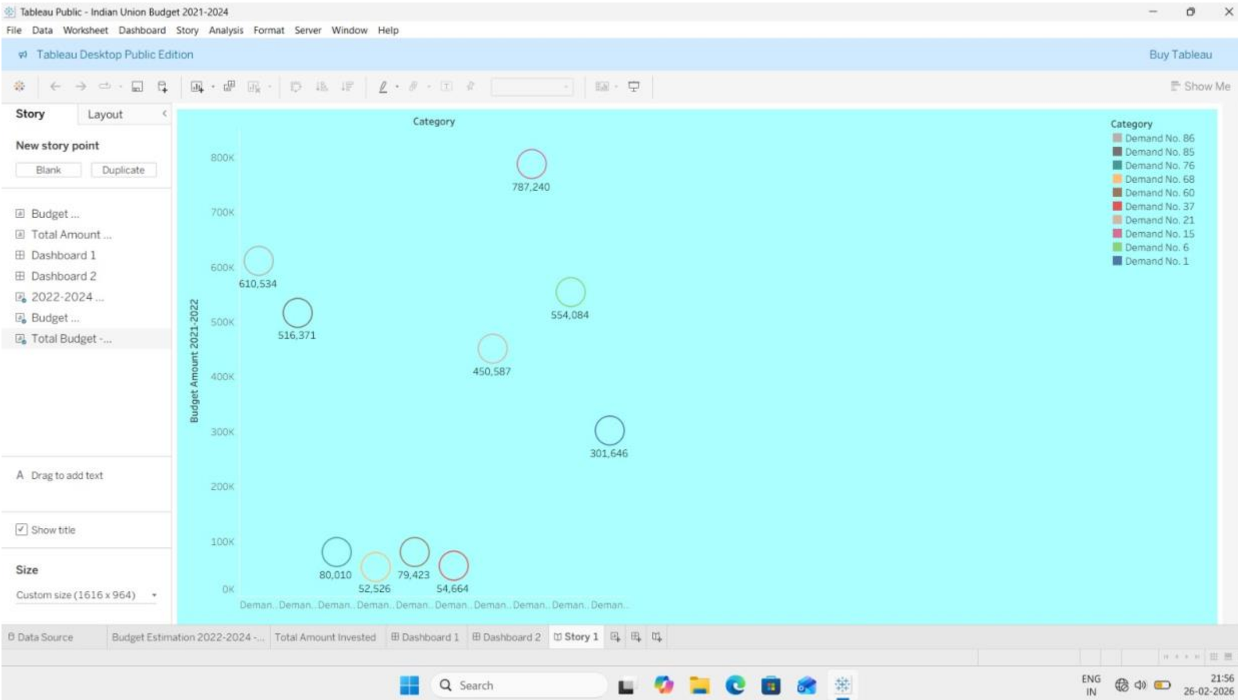
Data Source
Budget Estimation 2022-2024 -...
Total Amount Invested
Dashboard 1
Dashboard 2
Story 1



8.Advantages & Disadvantages

Advantages	Disadvantages
Trend Analysis – Understand historical budget patterns.	**Data Availability** – Some historical budgets may be incomplete.
Sector Prioritization – Identify high-priority sectors and allocations.	**Interpretation Bias** – Risk of subjective conclusions
Policy Support – Helps in planning and filling funding gaps	**Complexity** – Large financial datasets can be hard to manage
Socio-Economic Insights – Correlate allocation with development outcomes	**Accessibility** – Non-experts may struggle with technical data
Research & Education – Useful for economists, students, policymakers	**Time-Consuming** – Analysis of multiple years requires effort
Transparency – Promotes public awareness of government spending.	**Relevance** – Changing priorities may affect historical comparisons

9.Conclusion

Conclusion:

The analysis of Union Budget allocations over the years provides valuable insights into India's developmental priorities and government spending patterns. By examining trends across sectors, the project highlights areas of growth, funding gaps, and policy impact, enabling informed decision-making for policymakers, researchers, and citizens. While challenges like data complexity and changing priorities exist, the study ultimately promotes transparency, encourages evidence-based planning, and contributes to sustainable socio-economic development in India

10. Future Scope

- **Predictive Analysis** – Forecast future budget allocations and impacts.
- **Sector Studies** – Analyze critical sectors like health, education, and infrastructure.
- **Policy Assessment** – Evaluate effectiveness of government schemes.
- **Interactive Dashboards** – Provide dynamic tools for data exploration.
- **Public Awareness** – Educate citizens on budget trends and priorities
- **Comparative Studies** – Compare India's budget trends with other countries.
- **Sustainability Focus** – Identify allocations promoting long-term socio-economic growth.

11. Appendix

11.1 Dataset Link:

https://www.kaggle.com/datasets/prasenjitsharma/indian-union-budget-fy-21-22-till-23-24

11.2 Github & Project Demo link:

https://1drv.ms/v/c/a768281b984d8fbe/IQCM-mm0kv_kQ60PJpdOSdw_AQ7mCKdsAfGPy2_GqTRPwRs